

Classification Trees

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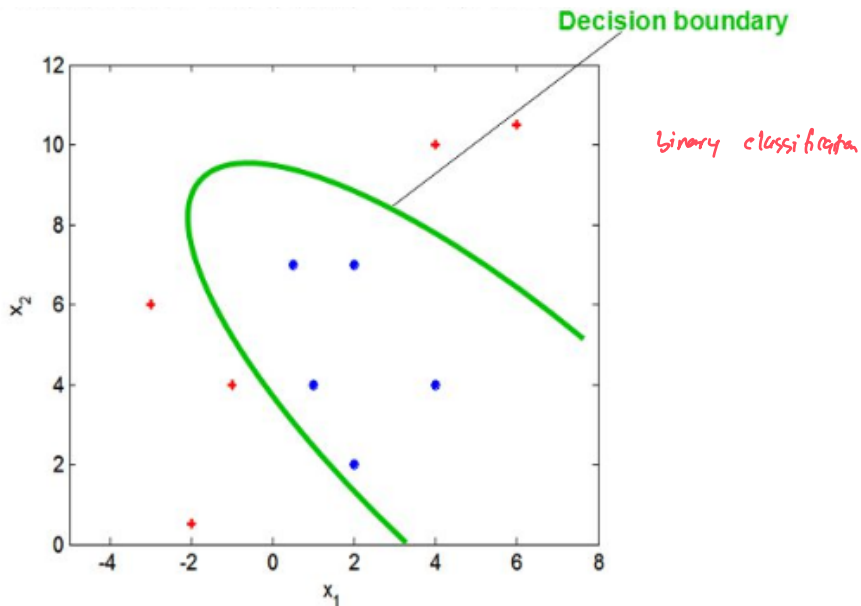
Reading Materials

- ▶ Max Kuhn. Chapter 14. Section 14.1

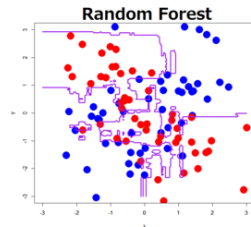
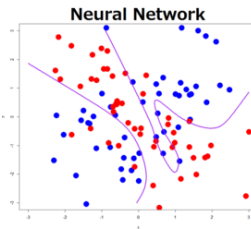
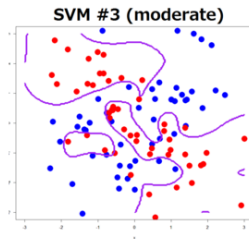
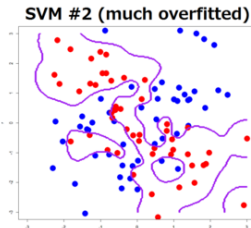
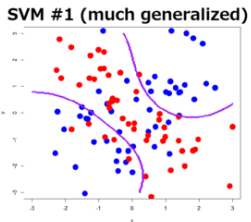
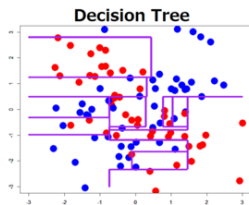
Decision Boundary in Classification

Classification is a process of finding the **decision boundary** that best separates two classes

Decision Boundary in Classification



Decision Boundary in Classification



► SVM = Support Vector Machine

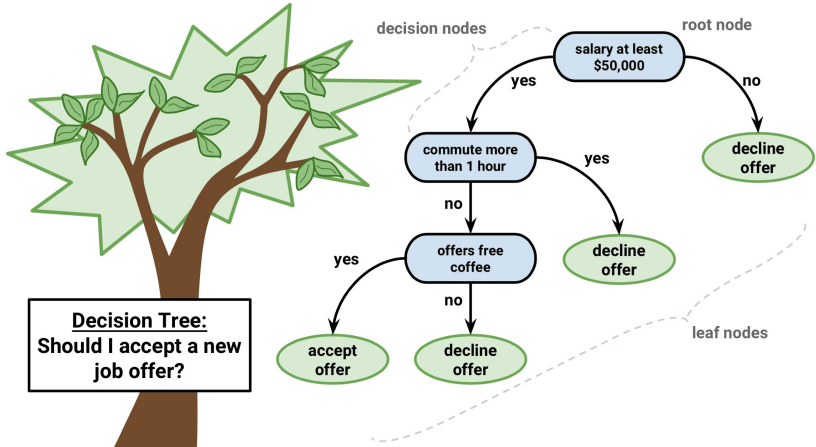
Decision Tree

- ▶ Decision Tree for classification is **Classification Tree**
- ▶ Decision Tree for Regression is **Regression Tree**

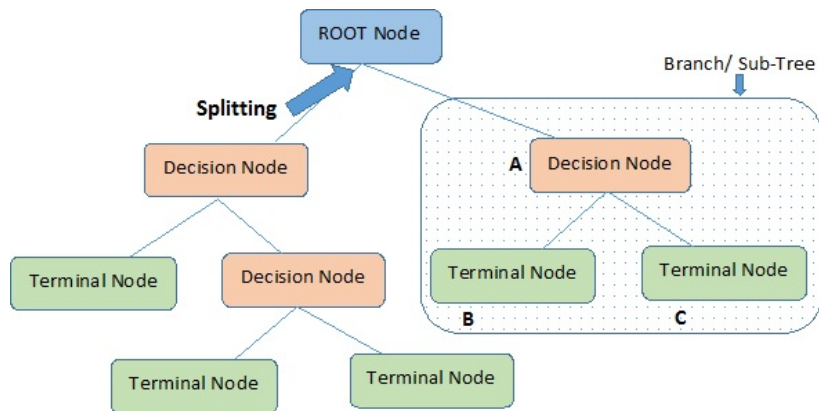
Example of Classification Tree

Link

Example of Classification Tree



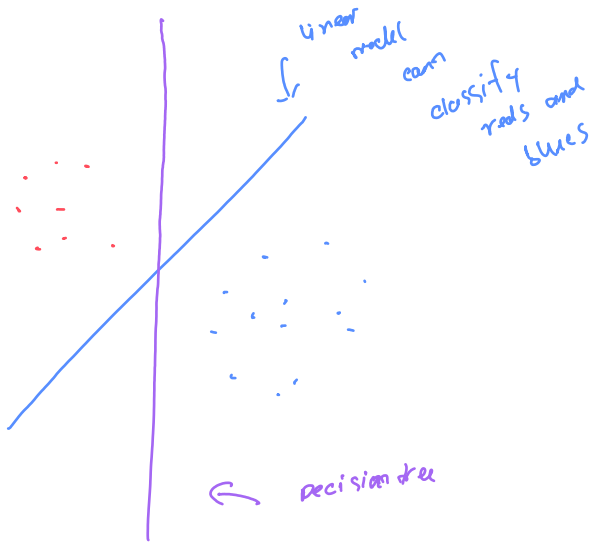
Example of Classification Tree



Note:- A is parent node of B and C.

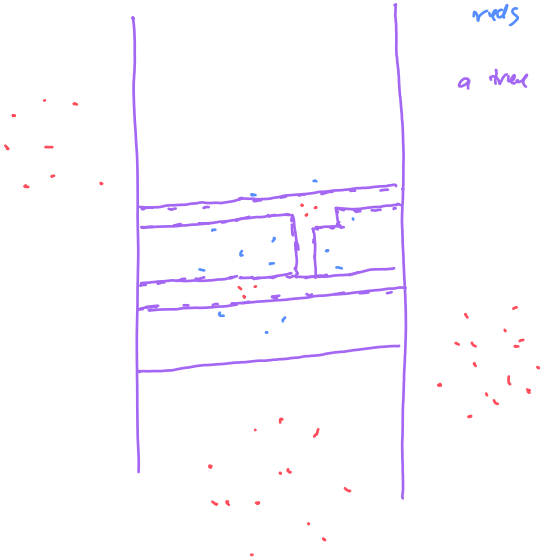
Classification Tree

- ▶ In two dimension, classification Tree's decision boundary is a collection of horiontal and vertical line

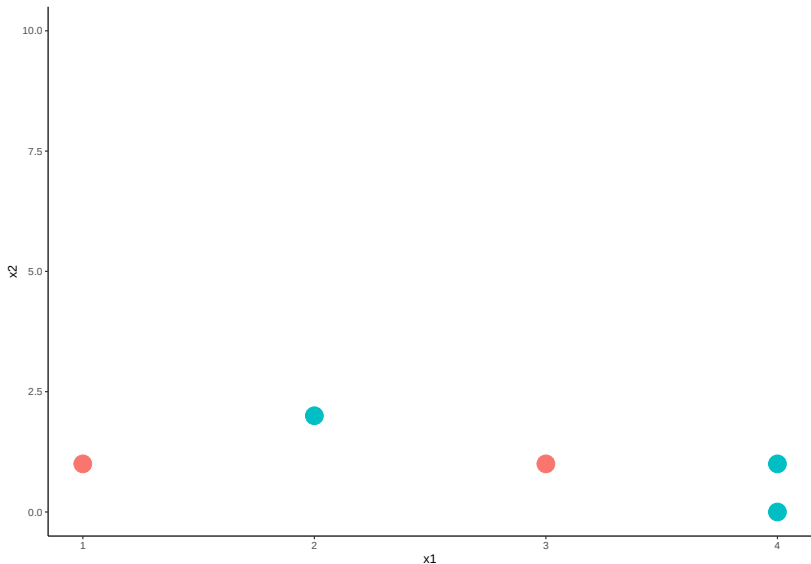


Linear model cannot classify
reds and blues

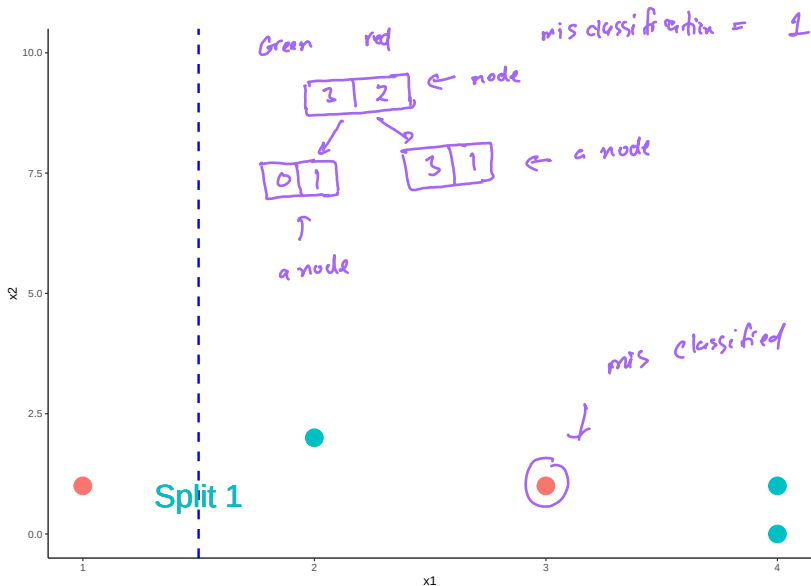
a tree can do that



Find a vertical line that best separate **red** and **green**.

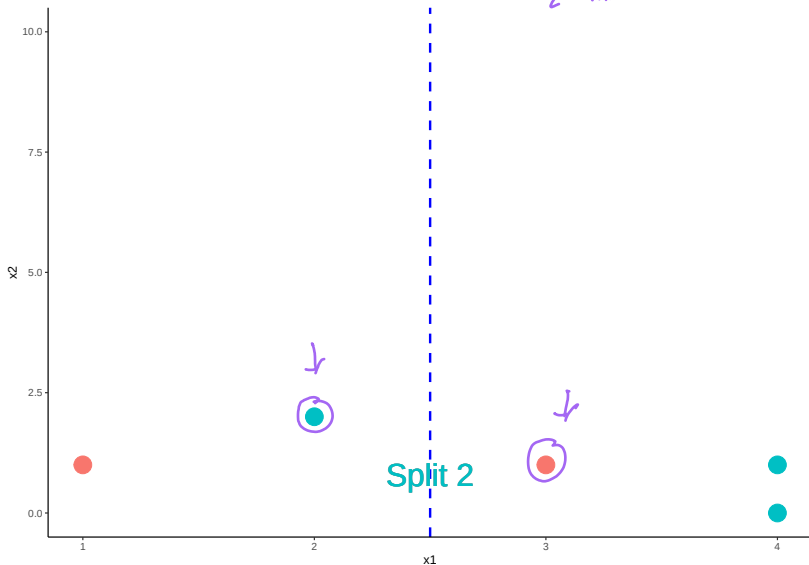


One way to separate the reds and greens



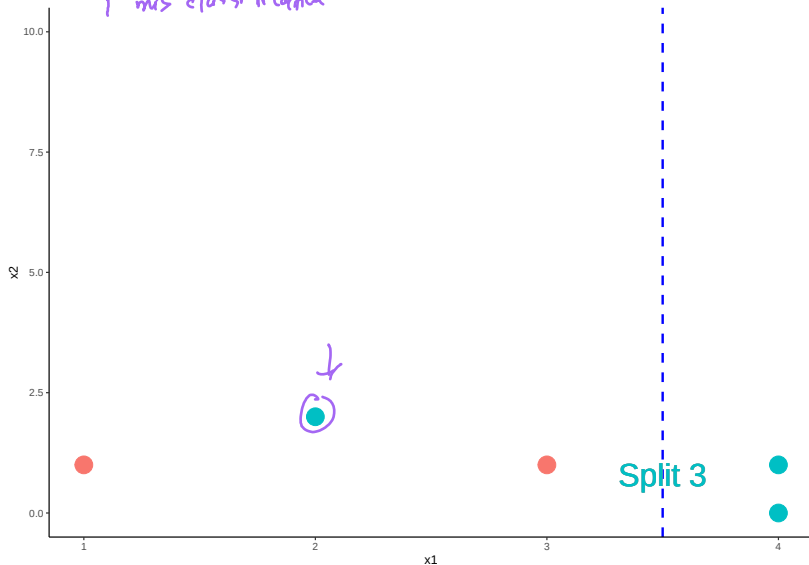
One way to separate the reds and greens

2 misclassification



One way to separate the reds and greens

1 mis classification



Question

- ▶ **Question:** Which is the best split?

Partial Answer

- ▶ It looks like Split 1 and 3 are better than Split 2 since it misclassifies less

Partial Answer

- ▶ Which is the better split between Split 1 and Split 3?

Partial Answer

- ▶ We need to find a way to measure *how good a split is*

Impurity Measure

- The impurity of a node (**a node = a subset of the data or the original data**) measure how **uncertain** the node is.

Impurity $\approx$ uncertain (unsure)

$$\text{Impurity} \left(\begin{array}{|c|c|} \hline 1 & 99 \\ \hline \end{array} \right) < \text{imp} \left(\begin{array}{|c|c|} \hline 40 & 60 \\ \hline \end{array} \right)$$

Impurity Measure

- ▶ For example, node A with 50% reds and 50% greens would be more uncertain than node B with 90% reds and 10% greens. Thus, node A has greater impurity than node B.

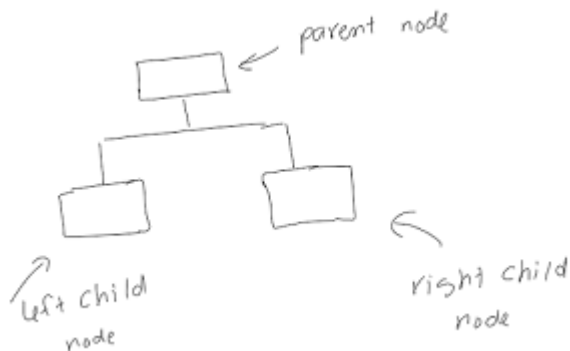
Impurity Measure

- ▶ More uncertain = Greater impurity

Children Impurity

- ▶ A split resulting smaller children impurity is a **better split**

Children Impurity ($I_{children}$)



$$I_{children} = \frac{N_{left}}{N} I_{left} + \frac{N_{right}}{N} I_{right}$$

- ▶ N_{left} and N_{right} are the number of points in the left child node and right child node, respectively.
- ▶ $N_{left} + N_{right} = N$

Impurity Measure

- ▶ Impurity can be measured by: classification error, Gini Index, and Entropy.

Impurity Measure

- Let p_0 and p_1 be the proportion of class 0 and class 1 in a node.

By Classification Error: $I = \min\{p_0, p_1\}$

By Gini Index: $I = 1 - p_0^2 - p_1^2$

By Entropy: $I = -p_0 \log_2(p_0) - p_1 \log_2(p_1)$

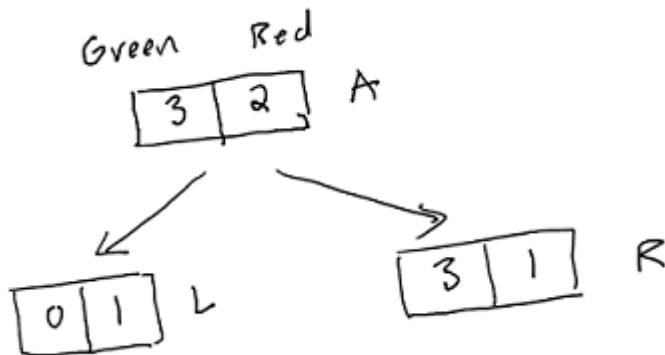
Calculation

- ▶ Let's calculate the Children Impurity ($I_{children}$) of the three splits to decide which split is the best

Split 1: Impurity by Classification Error

- ▶ Let **green** and **red** be class 0 and class 1, respectively.

For Split 1: $N = 5$, $N_{\text{left}} = 1$, $N_{\text{right}} = 4$



(*) By classification Error

(neg) 0 1 (pos)

p_0	p_1
-------	-------

 (node A)

$$I_A = \min \{ p_0, p_1 \}$$

p_0 is proportion of 0 (neg)

p_1 is proportion of 1 (pos)

Ex A

3	2
---	---

 $p_0 = \frac{3}{5}$; $p_1 = \frac{2}{5}$

$$\Rightarrow I_A = \min \left\{ \frac{3}{5}, \frac{2}{5} \right\} = \frac{2}{5}$$

(*) By Gini - Index

p_0	p_1
-------	-------

 A

$$I_A = 1 - p_0^2 - p_1^2$$

Ex: ①

3	2
---	---

 $\nearrow p_0 = 3/5$
 $\searrow p_1 = 2/5$

$$\Rightarrow I_A = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2$$

$$I_A = \underline{\underline{.48}}$$

②

3	3
---	---

node B

$$p_0 = p_1 = \frac{1}{2} \Rightarrow I_B = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2$$
$$= 0.5$$

$$\boxed{7} \quad \boxed{0}$$

node c

$$\Rightarrow p_0 = \frac{7}{7} = 1$$

$$p_1 = 0$$

$$\Rightarrow I_c = 1 - 1^2 - 0^2 = 0$$

Split 1



(we are using gini-index to measure impurity)

$$I_{\text{parent}} = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2 = .48$$

$$I_{\text{children}} = \frac{1}{5} \left[1 - 0^2 - 1^2 \right] + \frac{4}{5} \left[1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 \right]$$

$I_{\text{children}} = \frac{N_{\text{left}}}{N} I_{\text{left}} + \frac{N_{\text{right}}}{N} I_{\text{right}}$

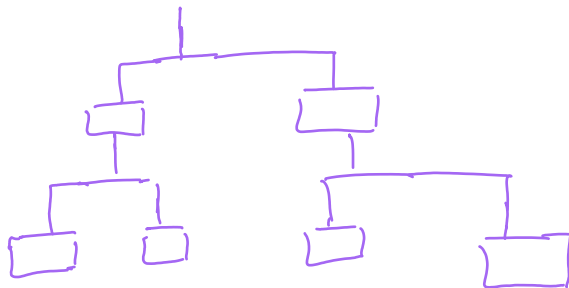
1 4
↖ ↖
↓ ↓
1 1
5

$$= \frac{4}{5} \cdot \frac{6}{16} = .3$$

$$I_{\text{gain}} = I_{\text{parent}} - I_{\text{children}} = .48 - .3 = .18$$

$$\begin{aligned} \underline{\text{split 2}} \quad I_{\text{gain}} &= .48 - .464 \\ &= .016 \end{aligned}$$

$$\begin{aligned} \underline{\text{split 3}} \quad I_{\text{gain}} &= .48 - .184 \\ &= .296 \end{aligned}$$



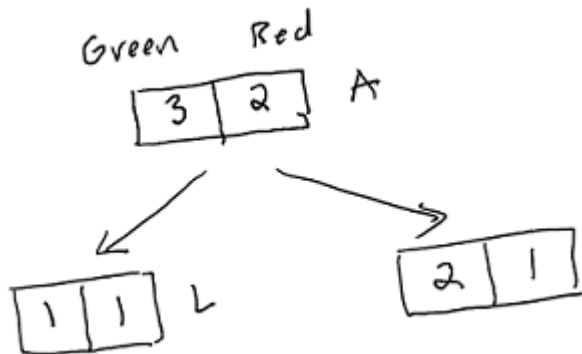
Split 1: Impurity by Classification Error

- ▶ Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus, $I_L = \min(0, 1) = 0$
- ▶ Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus, $I_R = \min(\frac{3}{4}, \frac{1}{4}) = \frac{1}{4}$
- ▶ Children Impurity of Split 1:

$$\begin{aligned} I_{children} &= \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R \\ &= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot \frac{1}{4} = 0.2 \end{aligned}$$

Split 2: Impurity by Classification Error

For Split 2: $N = 5$, $N_{\text{left}} = 2$, $N_{\text{right}} = 3$



Split 2: Impurity by Classification Error

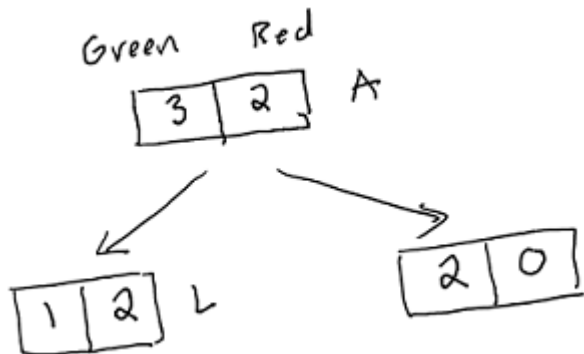
- ▶ Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus, $I_L = \frac{1}{2}$
- ▶ Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus, $I_R = \min(\frac{2}{3}, \frac{1}{3}) = \frac{1}{3}$
- ▶ Children Impurity of Split 2:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{2}{5} \cdot \frac{1}{2} + \frac{3}{5} \cdot \frac{1}{3} = 0.4$$

Split 3: Impurity by Classification Error

For Split 3: $N = 5$, $N_{\text{left}} = 3$, $N_{\text{right}} = 2$



Split 3: Impurity by Classification Error

- ▶ Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus, $I_A = \min(\frac{1}{3}, \frac{2}{3}) = \frac{1}{3}$
- ▶ Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus, $I_R = \min(1, 0) = 0$
- ▶ Children Impurity of Split 3:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{3}{5} \cdot \frac{1}{3} + \frac{2}{5} \cdot 0 = 0.2$$

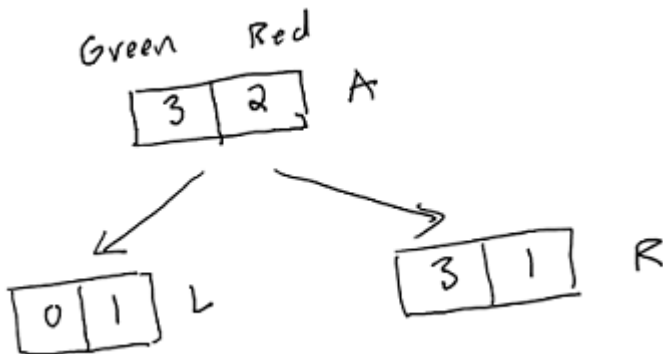
Comparing Impurity by Classification Error

	$I_{children}$
Split 1	0.2
Split 2	0.4
Split 3	0.2

- By classification error, Split 1 and Split 3 are tie as the best because they have the same Children Impurity ($I_{children}$).

Split 1: Impurity by Gini Index

For Split 1: $N = 5$, $N_{\text{left}} = 1$, $N_{\text{right}} = 4$



Split 1: Impurity by Gini Index

- ▶ Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus,

$$I_L = 1 - 0^2 - 1^2 = 0$$

- ▶ Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus,

$$I_R = 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 = 0.375$$

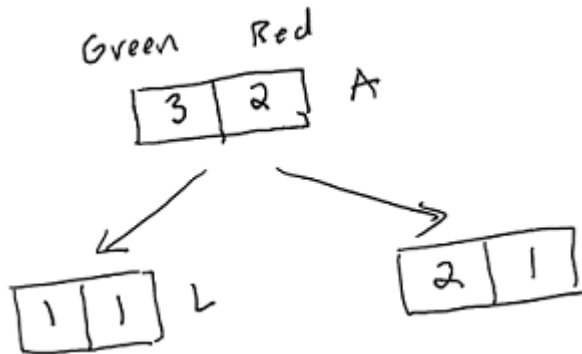
- ▶ Children Impurity of Split 1:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.375 = 0.3$$

Split 2: Impurity by Gini Index

For Split 2: $N = 5$, $N_{\text{left}} = 2$, $N_{\text{right}} = 3$



Split 2: Impurity by Gini Index

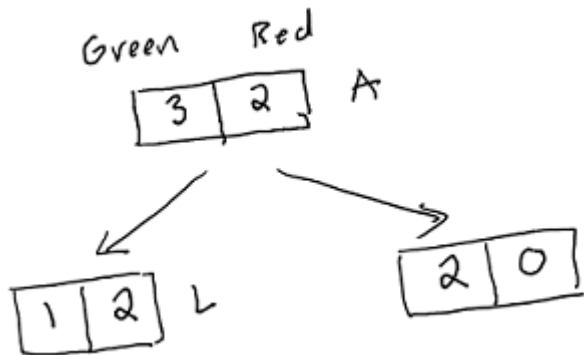
- ▶ Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus,
 $I_L = 1 - (\frac{1}{2})^2 - (\frac{1}{2})^2 = 0.5$
- ▶ Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus,
 $I_R = 1 - (\frac{2}{3})^2 - (\frac{1}{3})^2 = 0.44$
- ▶ Children Impurity of Split 2:

$$I_{children} = \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R$$

$$= \frac{2}{5} \cdot \frac{1}{2} + \frac{3}{5} \cdot 0.44 = \underline{0.464}$$

Split 3: Impurity by Gini Index

For Split 3: $N = 5$, $N_{\text{left}} = 3$, $N_{\text{right}} = 2$



Split 3: Impurity by Gini Index

- ▶ Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus,
 $I_A = 1 - (\frac{1}{3})^2 - (\frac{2}{3})^2 = 0.44$
- ▶ Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus,
 $I_R = 1 - 0^2 - 1^2 = 0$
- ▶ Children Impurity of Split 3:

$$\begin{aligned} I_{children} &= \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R \\ &= \frac{3}{5} \cdot 0.44 + \frac{2}{5} \cdot 0 = 0.184 \end{aligned}$$

Comparing Impurity by Gini Index

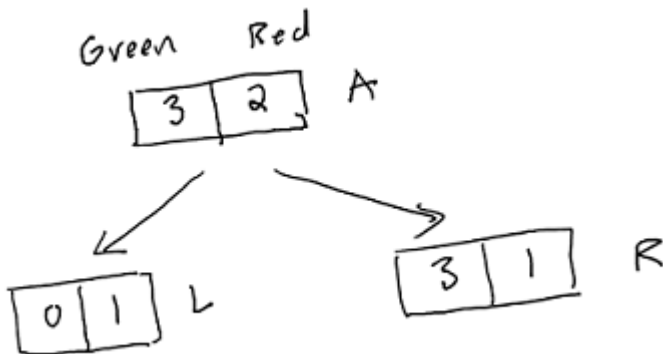
	$I_{children}$
Split 1	0.3
Split 2	0.464
Split 3	0.184

← best split
(lowest impurity)

- By Gini Index, Split 3 is the best because it has the smallest Children Impurity ($I_{children}$).

Split 1: Impurity by Entropy

For Split 1: $N = 5$, $N_{\text{left}} = 1$, $N_{\text{right}} = 4$



Split 1: Impurity by Entropy

- ▶ Node *child left*, L: $p_0 = \frac{0}{1} = 0, p_1 = \frac{1}{1} = 1$. Thus, $I_L = 0$
- ▶ Node *child right*, R: $p_0 = \frac{3}{4}, p_1 = \frac{1}{4}$. Thus,

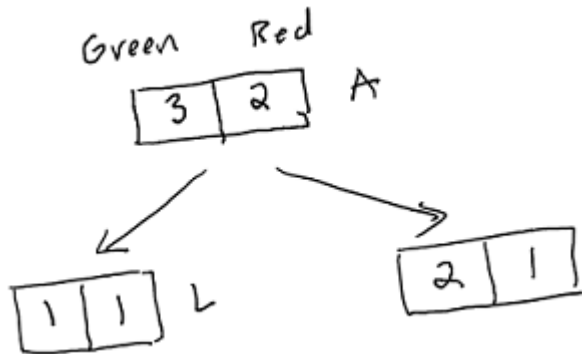
$$I_R = -\log_2\left(\frac{3}{4}\right) - \log_2\left(\frac{1}{4}\right) = 0.811$$

- ▶ Children Impurity of Split 1:

$$\begin{aligned} I_{\text{children}} &= \frac{N_{\text{left}}}{N} I_L + \frac{N_{\text{right}}}{N} I_R \\ &= \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.811 = 0.649 \end{aligned}$$

Split 2: Impurity by Entropy

For Split 2: $N = 5$, $N_{\text{left}} = 2$, $N_{\text{right}} = 3$



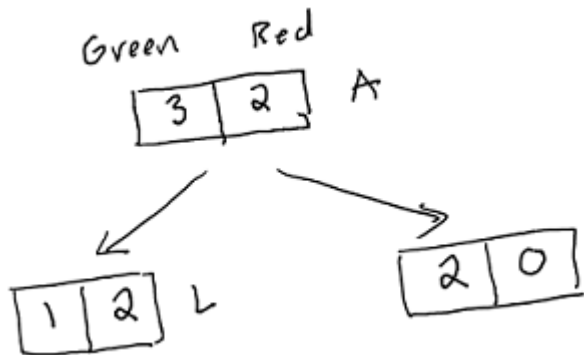
Split 2: Impurity by Entropy

- ▶ Node *child left*, L: $p_0 = \frac{1}{2}, p_1 = \frac{1}{2}$. Thus,
 $I_L = -\log_2(\frac{1}{2}) - \log_2(\frac{1}{2}) = 1$
- ▶ Node *child right*, R: $p_0 = \frac{2}{3}, p_1 = \frac{1}{3}$. Thus,
 $I_R = -\log_2(\frac{2}{3}) - \log_2(\frac{1}{3}) = 0.918$
- ▶ Children Impurity of Split 2:

$$\begin{aligned}I_{children} &= \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R \\&= \frac{2}{5} \cdot 1 + \frac{3}{5} \cdot 0.918 = 0.951\end{aligned}$$

Split 3: Impurity by Entropy

For Split 3: $N = 5$, $N_{\text{left}} = 3$, $N_{\text{right}} = 2$



Split 3: Impurity by Entropy

- ▶ Node *child left*, L: $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$. Thus,
 $I_A = -\log_2(\frac{1}{3}) - \log_2(\frac{2}{3}) = 0.918$
- ▶ Node *child right*, R: $p_0 = \frac{2}{2}, p_1 = \frac{0}{2}$. Thus, $I_R = 0$
- ▶ Children Impurity of Split 3:

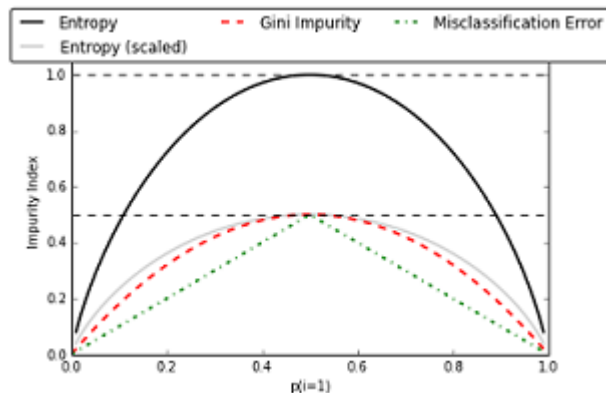
$$\begin{aligned}I_{children} &= \frac{N_{left}}{N} I_L + \frac{N_{right}}{N} I_R \\&= \frac{3}{5} \cdot 0.918 + \frac{2}{5} \cdot 0 = 0.551\end{aligned}$$

Comparing Impurity by Entropy

	$I_{children}$
Split 1	0.649
Split 2	0.951
Split 3	0.551

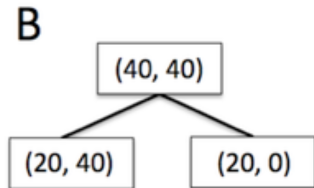
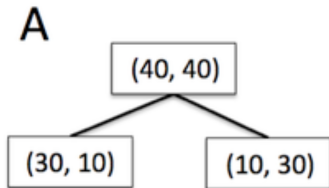
- By Gini Index, Split 3 is the best because it has the smallest Children Impurity ($I_{children}$).

Comparing Impurity Measures



- Relation between impurity and the class probabilities. All impurity measures are maximized at $p_1 = 1/2$ and minimized at $p_1 = 0$ and $p_1 = 1$.

Another Example (submit to convos for extra credit)
calculating using entropy or Gini-index
or classification errors)



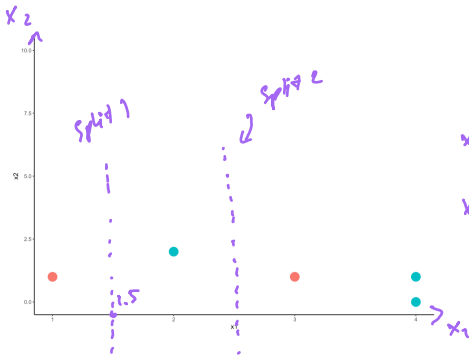
- Which split is better?

visually, which one is better?

A better : 8

B better : 1

same : 0



split 1

	Red	Green
$x_1 \geq 1.5$	1	3
$x_1 < 1.5$	1	0

⊕ **Variable 1** (binary variable)

- Red
- Green

⊕ **Variable 2** (binary)

- yes ($x_1 \geq 1.5$)
- no (not ≥ 1.5)

split 2

	Red	Green
$x_1 \geq 2.5$	1	2
$x_1 < 2.5$	1	1

} v_4

Fav. music
(X)

Gender
(Y)

Test:

H_0 : X and Y are independent

H_2 : X and Y are dependent

→ collect data → perform a Chi-squared test

↗ (1) χ^2 -test statistic. ↗ small → support H_0
↘ large → support H_2

↘ (2) p-value ↗ "small" ($\leq .05$) → support H_2
↘ "large" ($> .05$) → support H_0

Decide the best split using Chi-Square test of Independence

- ▶ Besides Children Impurity, one can use the Chi-square, χ^2 , test of independence to decide the best split.

Review of Chi-Square test of Independence

- ▶ Let X and Y be two categorical variables.
- ▶ We want to test if X and Y are independent/associated
 - ▶ H_0 : X and Y are independent
 - ▶ H_a : X and Y are dependent
- ▶ Test statistic:

$$\sum \frac{(e_i - o_i)^2}{e_i} \sim \chi^2 \text{ distribution with degree of freedom } (n-1)(m-1)$$

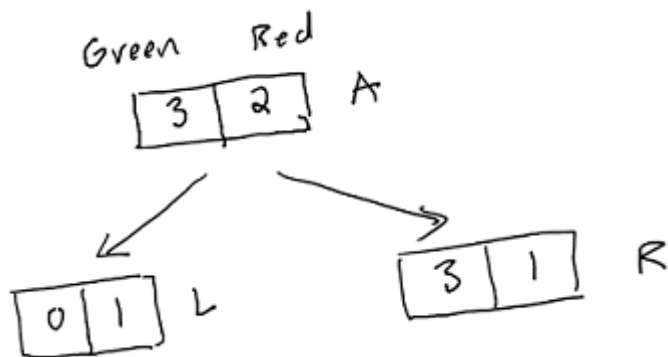
Review of Chi-Square test of Independence

- ▶ In our context, the greater the χ^2 value, the smaller the *p – value*
- ▶ The smaller the *p – value*, the more dependent the two variables are. Thus the better the split is.
- ▶ Therefore, we look for the split with the **greatest χ^2 value.**

Applying to Our Example

- ▶ We will calculate the χ^2 values of the three splits.
- ▶ The best split is the split with the greatest χ^2 value.

Split 1



	Greens	Reds	Total
Left Branch	0	1	1
Right Branch	3	1	4
Total	3	2	

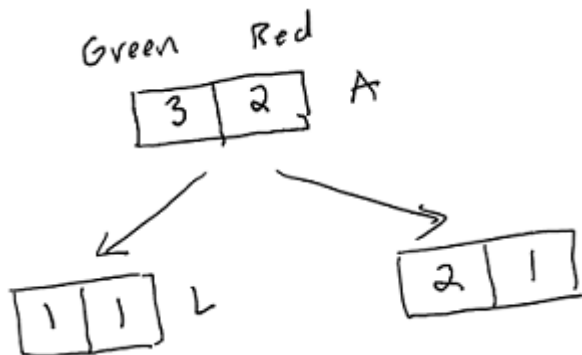
Split 1

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- ▶ $i = 1$ (Cell 1): $e_1 = \frac{1 \cdot 3}{5}$, $o_1 = 0$
- ▶ $i = 2$ (Cell 2): $e_2 = \frac{1 \cdot 2}{5}$, $o_2 = 1$
- ▶ $i = 3$ (Cell 3): $e_3 = \frac{3 \cdot 4}{5}$, $o_3 = 3$
- ▶ $i = 4$ (Cell 4): $e_4 = \frac{2 \cdot 4}{5}$, $o_4 = 1$
- ▶ Plug in, we have:

$$\chi^2 = 1.875$$

Split 2



	Greens	Reds	Total
Left Branch	1 (Cell 1)	1 (Cell 2)	2
Right Branch	2 (Cell 3)	1 (Cell 4)	3
Total	3	2	

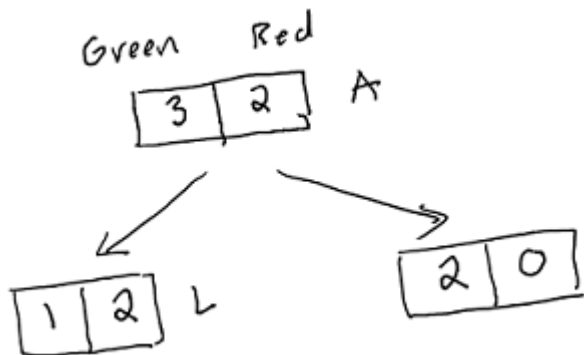
Split 2

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- ▶ $i = 1$ (Cell 1): $e_1 = \frac{2 \cdot 3}{5}$, $o_1 = 1$
- ▶ $i = 2$ (Cell 2): $e_2 = \frac{2 \cdot 2}{5}$, $o_2 = 1$
- ▶ $i = 3$ (Cell 3): $e_3 = \frac{3 \cdot 3}{5}$, $o_3 = 2$
- ▶ $i = 4$ (Cell 4): $e_4 = \frac{3 \cdot 2}{5}$, $o_4 = 1$
- ▶ Plug in, we have:

$$\chi^2 = 0.139$$

Split 3



	Greens	Reds	Total
Left Branch	1 (Cell 1)	2 (Cell 2)	3
Right Branch	2 (Cell 3)	0 (Cell 4)	2
Total	3	2	

Split 3

$$\chi^2 = \frac{(e_1 - o_1)^2}{e_1} + \frac{(e_2 - o_2)^2}{e_2} + \frac{(e_3 - o_3)^2}{e_3} + \frac{(e_4 - o_4)^2}{e_4}$$

- ▶ (Cell 1): $e_1 = \frac{2 \cdot 3}{5}$, $o_1 = 1$
- ▶ (Cell 2): $e_2 = \frac{2 \cdot 2}{5}$, $o_2 = 2$
- ▶ (Cell 3): $e_3 = \frac{3 \cdot 3}{5}$, $o_3 = 2$
- ▶ (Cell 4): $e_4 = \frac{3 \cdot 2}{5}$, $o_4 = 0$
- ▶ Plug in, we have:

$$\chi^2 = 2.222$$

Comparing the three splits

	χ^2
Split 1	1.875
Split 2	0.139
Split 3	2.222

- Split 3 is the best because it has the greatest χ^2 !

Logworth

- ▶ The quality of the split can be measured by **Logworth**
- ▶ Formula:

$$\text{logworth} = -\log(p_{\text{value}})$$

- ▶ The greater the logworth, the better the split

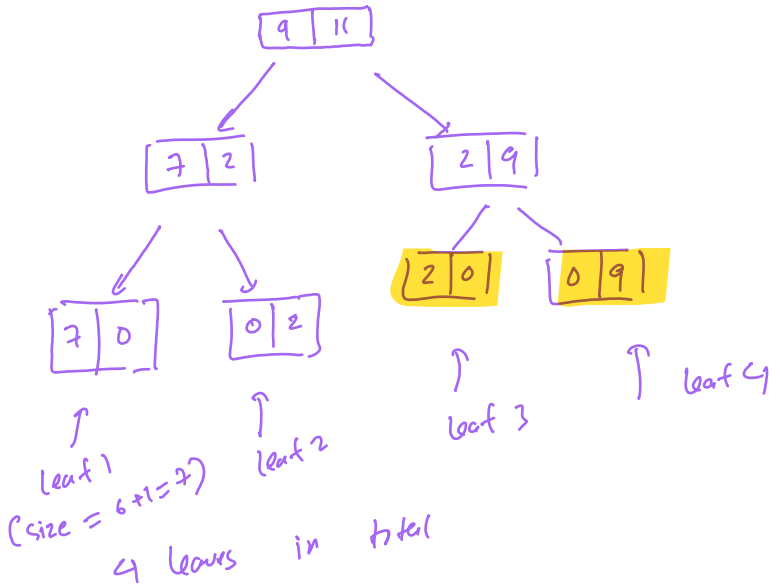
Logworth

	χ^2	p-value	logworth
Split 1	1.875	0.114	0.943
Split 2	0.139	0.998	0.0008
Split 3	2.222	0.088	1.055

$= -\log(p\text{ value})$

- ▶ Greatest χ^2 = Lowest p - value = Greatest logworth = Best Split
- ▶ Split 3 is the best split!

Growing a tree



What happens after the first split?

- ▶ After the first split, the data are divided into to subsets.
- ▶ The splitting process is repeated for each subset.
- ▶ The process ends when a stopping criteria is satisfied

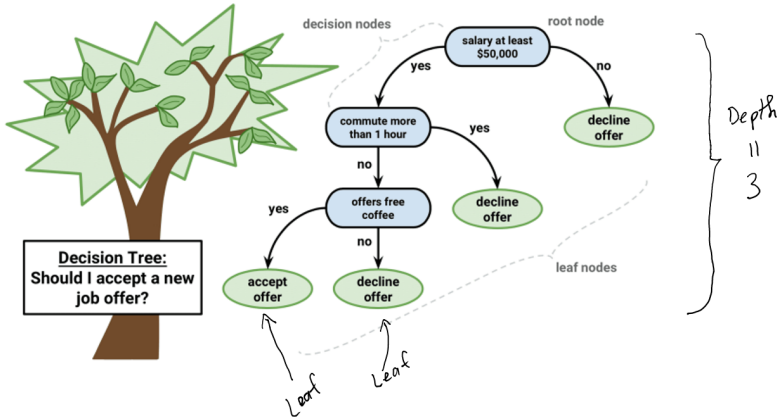
Example ; depth is 5

OK ; number of leaves : 20

Stopping Criteria

- ▶ Minimum Leaf Size: The minimum of observations in the leaves
- ▶ Maximum Number of Leaves
- ▶ Maximum Depth
- ▶ Others

Stopping Criteria



Decision Tree Algorithm - How to grow a tree

- ▶ Step 1: Calculate the Children Impurity or p – *value* of all possible splits at all variables
- ▶ Step 2: Select the split that give the minimum Children Impurity or lowest p – *value* to split the data into two subdata D_1 and D_2
- ▶ Repeat *Step 1* and *Step 2* to both D_1 and D_2 .
- ▶ Until a stopping criteria is satisfied

Complexity of Decision Tree

- ▶ A complexity of a tree can be measured by the number of leaves the tree has
- ▶ The more leaves a tree has, the more complex the tree is.
- ▶ A complex tree may be **overfitted**, i.e. having low training error but high testing error.

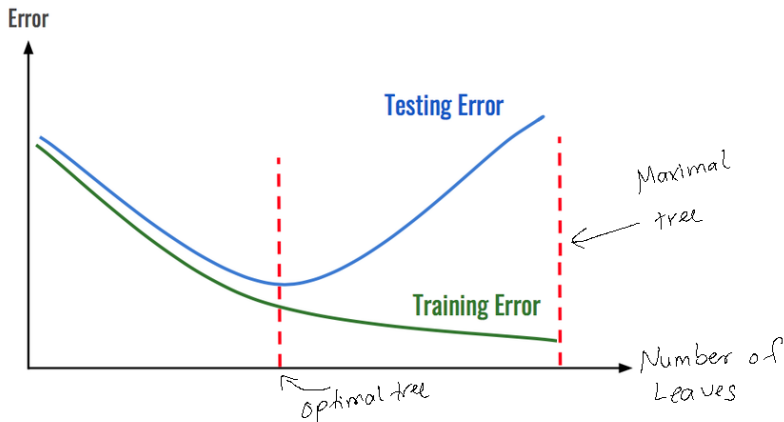
Pruning a tree

- ▶ For any given data, one can construct a tree that achieves 0 misclassification on training data
- ▶ After growing the tree one needs to prune it to avoid overfitted

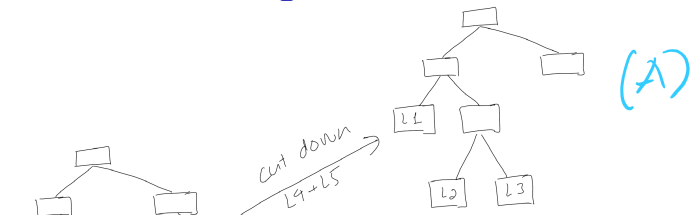
Pruning a tree

- ▶ The tree with maximum number of leaves is called the **maximal tree** (still satisfied the stopping rule)
- ▶ From the **maximal tree**, leaves are cut down, one by one, to obtain all possible subtrees
- ▶ The subtree with lowest error on validation data, is the **optimal tree**

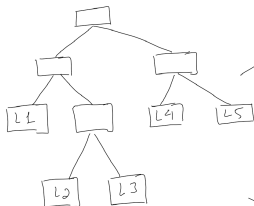
Maximal vs Optimal Tree



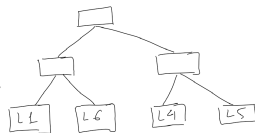
Example of Tree Pruning



(A)



Cut down
 $L2 + L3$



(B)

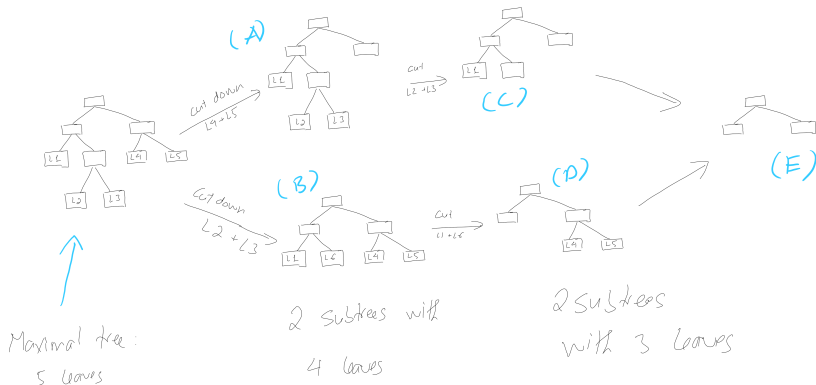
Maximal tree:

5 leaves

2 subtrees with

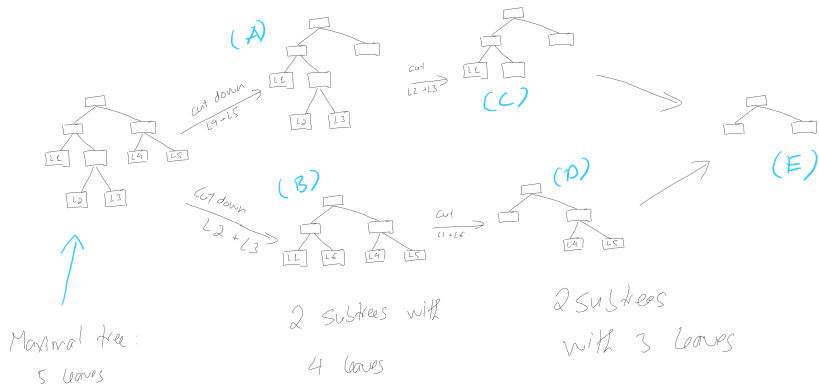
4 leaves

Example of Tree Pruning



- ▶ All the subtrees A, B, C, D, and E will be validated with the validation data to find the **optimal tree**
- ▶ The **optimal tree** could be the **maximal tree**!

Question



- What if both B and C give the lowest error on the validation data? Which tree should be selected as the final model?

we always choose the simplest model with the best performance

⇒ select C

impute missing data

①

Sex
F
F
M
F
F
M
M

F → Missed ← replace by the majority

F → missing ← which is F

②

Age
20
30
10
40
Miss
Miss

25 → Miss ← replace
25 → Miss by average
= 25

Train a model in Python

Step 1: Import + clean data

Step 2: handle missing data

Step 3: Encoding categorical variables  no need to do

Step 4: Train + Tune models

Step 5: Score on new data
predict

(*) Encoding categorical variables

!!

convert categorical → numeric

Fav. music
Pop
Rock
RB
country
others
RB
Pop

Encode
→
(not good)

Fav. music
1
2
3
4
5
3
1

↓ Encode (dummy encoding or 1 hot encoding)

Fav. music
Pop
Rock
RB
country
others
RB
Pop

(better)
→

	Pop	Rock	RB	Country	others
Pop	1	0	0	0	0
Rock	0	1	0	0	0
RB	0	0	1	0	0
country	0	0	0	1	0
others	0	0	0	0	1
RB	0	0	1	0	0
Pop	1	0	0	0	0

